IDENTIFYING METALLIC ELEMENTS

When burnt in a flame, many metallic elements burn with a distinctive colour. In this experiment, you will burn a number of metallic salts in a Bunsen burner flame and record the colour of each element.

Each element will be set up at a different Bunsen burner around the room. You will need to rotate through each station and test all elements.

AIM:

METHOD

1. For each salt, write the symbol of the metal present. Use your periodic table to help you.
2. Write the ions of as many metals as you can, using your periodic table.
3. At each station, carefully spray the solution into the flame
4. Record the colour of the flame in the table

RESULTS

Substance Tested	Symbol of metal	Metal Ion Present	Flame Colour
Barium chloride			
Sodium chloride			
Magnesium chloride			
Calcium chloride			
Potassium chloride			
Lithium chloride			
Copper chloride			
Lead nitrate			

QUESTIONS

1. Use your results table to identify the two unknowns.

	Flame Colour	Metal Ion Present
Unknown 1		
Unknown 2		

2. Why should you only use one salt at each Bunsen burner?

3. Why should you use safety glasses in this experiment?

4. How is the flame test useful to scientists?

5. Where else might this property of metals be useful?

6. Use your periodic table to predict the ions that will form from the following elements.

Element	Symbol	Loses or Gains Electrons	Ion Formed
Oxygen	O	Gains electrons	O^{-2}
Sodium			
Nitrogen			
Calcium			
Aluminium			
Fluorine			